CURRICULUM VITAE

# Bo-Yu Chen 陳柏昱

Researcher and builder working across music AI, creative tools, musical interfaces, and human–computer interaction.

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## RESEARCH INTERESTS

Human-Computer Interaction (HCI) · Music Information Retrieval (MIR) · Tools for Thought

## EDUCATION

**National Taiwan University (NTU)** Sep 2021 – Aug 2025
M.S. in Networking and Multimedia (Dept. of CSIE) · GPA 4.21 / 4.3 Taipei, Taiwan

- **Master's Thesis:** *FlueBricks* [C1] (Advisor: Prof. Lung-Pan Cheng)
- TAICHI 2025 Best Paper | TAICHI 2026 Thesis Honorable Mention

**National Taiwan University of Science and Technology (NTUST)** Sep 2013 – Jun 2018
B.S. in Electrical Engineering Taipei, Taiwan

## RESEARCH EXPERIENCE

**Computational Physicality Lab, National Taiwan University** Sep 2021 – Aug 2025
Graduate Research Assistant (Advisor: Prof. Lung-Pan Cheng) Taipei, Taiwan

- **Conceptualized acoustic reasoning through FlueBricks**, a modular construction kit connecting instrument design and play in a **designer–player loop**. [C1] TAICHI 2025 Best Paper TAICHI 2026 Thesis Honorable Mention
- **Engineered parametric CAD assemblies in Fusion 360** and conducted a **12-participant study** spanning novices to professional musicians, exploring acoustic understanding and musical expression.

**Creative AI Lab, Sony Group Corporation** Nov 2020 – Sep 2021
Research Intern (Advisors: Dr. Yuki Mitsufuji, Dr. Marco A. Martínez Ramírez, Wei-Hsiang Liao) Tokyo, Japan (Remote)

- **Developed DJtransGAN**, the first generative model for automated cross-track DJ transitions using DDSP and GANs. [C2]
- **Open-sourced DJtransGAN ( ) and its data-generation pipeline ( )**, earning **120+ GitHub stars** and community interest.

**Music and AI Lab, Academia Sinica** Jul 2019 – Sep 2023
Research Assistant (Advisor: Prof. Yi-Hsuan Yang) Taipei, Taiwan

- **Pioneered loop-based music AI research** spanning compatibility assessment, generation, and benchmarking. [C3] [C4] [C5]
- **Mentored 5+ junior researchers**, guiding two to first-author ISMIR papers. [C3] [C4]

## WORK EXPERIENCE

**Rhythm Culture Corporation** Jul 2023 – Present
Technical Lead Taipei, Taiwan

- **Architecting Rytho**, a local-first creative workspace that enables AI agents to listen to, understand, search for, and create music alongside people.
- **Led R&D on LLM-driven music agents**, developing conversational feedback agents for music production and reasoning systems for chord analysis. [W1] [W2]
- **Managed engineering teams for government R&D projects** (TAICCA, Taipei City) and **TMC-CL1 ( )**, a web music interface commissioned by Taipei Music Center.

**Zone Sound Creative Corporation** Mar 2019 – Aug 2019
Interactive Engineer Taipei, Taiwan

- **Architected interactive web audio interfaces and networked performance systems**, developing a vocal sampler for Vocal Asia (Japan) [A2] and a p5.js/Node.js WebSocket engine bridging mobile audiences to Max/MSP stage synthesis [A4].
- **Developed a web demo showcasing Taiwanese electronic music** in collaboration with electronic music artists for MIDEM (Cannes, France). [A3]

## PUBLICATIONS

C = Conference papers · W = Workshop papers

**C1** **FlueBricks: A Construction Kit of Flute-like Instruments for Acoustic Reasoning**
**Bo-Yu Chen**, Chiao-Wei Huang, Lung-Pan Cheng
ACM Conference on Human Factors in Computing Systems (**CHI 2026**)
Acceptance ~25% · TAICHI 2025 · Best Paper · DOI: 10.1145/3772318.3790595

**W1** **Enhancing Automatic Chord Recognition through LLM Chain-of-Thought Reasoning**
Chih-Cheng Chang, **Bo-Yu Chen**, Lu-Rong Chen, Li Su
International Society for Music Information Retrieval Conference (**ISMIR 2025**) – LLM4Music Workshop
arXiv: 2509.18700

**W2** **AI TrackMate: Finally, Someone Who Will Give Your Music More Than Just "Sounds Great!"**
Yi-Lin Jiang, Chia-Ho Hsiung, Yen-Tung Yeh, Lu-Rong Chen, **Bo-Yu Chen**
Conference on Neural Information Processing Systems (**NeurIPS 2024**) – Creative AI Track
Acceptance 31% · arXiv: 2412.06617

**C2** **Automatic DJ Transitions with Differentiable Audio Effects and Generative Adversarial Networks**
**Bo-Yu Chen**, Wei-Han Hsu, Wei-Hsiang Liao, Marco A. Martínez Ramírez, Yuki Mitsufuji, Yi-Hsuan Yang
IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP 2022**)
Acceptance 47% · DOI: 10.1109/ICASSP43922.2022.9746663

**C3** **Exploiting Pre-trained Feature Networks for Generative Adversarial Networks in Audio-Domain Loop Generation**
Yen-Tung Yeh, **Bo-Yu Chen**, Yi-Hsuan Yang
International Society for Music Information Retrieval Conference (**ISMIR 2022**)
Acceptance 43% · DOI: 10.5281/zenodo.7316614

**C4** **A Benchmarking Initiative for Audio-Domain Music Generation Using the Freesound Loop Dataset**
Tun-Min Hung, **Bo-Yu Chen**, Yen-Tung Yeh, Yi-Hsuan Yang
International Society for Music Information Retrieval Conference (**ISMIR 2021**)
Acceptance 34% · DOI: 10.5281/zenodo.5624409

**C5** **Neural Loop Combiner: Neural Network Models for Assessing the Compatibility of Loops**
**Bo-Yu Chen**, Jordan Smith, Yi-Hsuan Yang
International Society for Music Information Retrieval Conference (**ISMIR 2020**)
Acceptance 38% · DOI: 10.5281/zenodo.4245462

## HONORS & AWARDS

**Master's Thesis Honorable Mention Award** · TAICHI Thesis Award for *FlueBricks* (published as [C1]) Top 16.6% — 2026

**Best Paper Award** 🏆 · Taiwan Computer-Human Interaction Society (TAICHI) for *FlueBricks* [C1] — 2025

**Overseas Visit Delegate** · Epoch Foundation Young Entrepreneurs of the Future Top 9% · 16/175 — 2017

**3rd Prize** · Ministry of Economic Affairs (MOEA) GetFresh Creativity Competition — 2017

## TEACHING & SERVICE

**BLND Creative Coding School** — Aug 2020 – Jul 2023
Instructor — Taipei, Taiwan

- Taught generative AI art (GANs, Stable Diffusion workflows) and physical computing with Arduino to creative practitioners.

**OrangeApple Children Programming School** — Jul 2015 – Sep 2019
Instructor — Taipei, Taiwan

- Taught algorithms and C++/JS to K-12 students; coached high school students for the APCS examination.

### ACADEMIC SERVICE & REVIEWING

International Society for Music Information Retrieval Conference (**ISMIR**) 2020 – 2026

IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2025

Conference on Neural Information Processing Systems (**NeurIPS**) – Creative AI Track 2025

## INTERACTIVE ART & INSTALLATIONS

As the interactive engineer on these works, I built the web front-ends, real-time audio, networked performance systems, and Arduino-based physical interaction, in collaboration with artists, musicians, and designers.

**A1** **Iridescent Vision** 2019
A web-based interactive music video and light installation exploring human civilization and imagined future technology.
*With DJ Sonia Calico and visual artist Celso Urroz · Presented at FINAL Club, Taipei, Taiwan*

**A2** **Taptap** 2019
A web-based interface that lets visitors play music with a cappella samples.
*With O-KAI SINGERS · Presented at Vocal Asia Festival, Kagawa, Japan*

**A3** **FusionTaiwanBeat** 2019
A web demo showcasing Taiwanese electronic music.
*With DJ Sonia Calico and visual artist Celso Urroz · Presented at MIDEM, Cannes, France*

**A4** **Duplex Transmission** 2019
A web-based system that connects audiences and performers during live performance.
*With new media artist Po-Hao Chi · Presented at Songshan Cultural Park, Taipei, Taiwan*

**A5** **Mimo No.Tent** 2019
A virtual planetarium that brings a starry sky inside a tent.
*With designers Yi-Yun Liao and Jing-Ting Huang · Presented at TaiwanTech Design Grad Show, Taipei, Taiwan*

**A6** **YuYu** 2018
An AR service that delivers customized messages through Chinese pastries.
*With designer Ya-Cin Siao · Presented at TaiwanTech Design Grad Show, Taipei, Taiwan*

**A7** **Message Container** 2017
A breath-driven installation that responds with haptic, sound, and visual feedback.
*With new media artist Wen-Fei Zhang · Presented at TaiwanTech Design Grad Show & Venue, Taipei, Taiwan*

## OPEN-SOURCE SOFTWARE & TOOLKITS

**OpenMIRLab** Nov 2025 – Present
Founder & Lead Maintainer Open-Source

- **Building OpenMIRLab to make music AI research easier to use and build on**, bringing 30+ MIR models into a consistent, easy-to-install toolkit.
- **Standardized unified inference APIs and checkpoint registries**, stripping heavy training dependencies into lightweight, pip-installable packages with verified model outputs.

## EXTRACURRICULAR ACTIVITIES

**Musical Theater: Rose of the Endless Night** Feb 2015 – Jun 2015
Director, Music Theater Course Final Public Performance (NTUST) Taipei, Taiwan

- Adapted *Rose of the Endless Night* for the stage and directed 60+ cast and crew members for an audience of 400+.

**NTUST Men's Football Team** Jul 2016 – Jul 2017
Vice Captain, Position: Defensive Midfielder Taipei, Taiwan

- Coordinated squad training and match operations across 30 collegiate athletes and coaching staff.
- Led team to **4th Place** in the National Inter-University Football Tournament and **5th Place** in the National Futsal Tournament.